Levers

LAB LESSON

OBJECTIVE

In this lesson, students will learn about the three main types of levers and how the arrangement of their components differ.

ACARA CONTENT DESCRIPTIONS

Change to an object's motion is caused by unbalanced forces, including Earth's gravitational attraction, acting on the object (ACSSU117)

- investigating a simple machine such as a lever

Planning and conducting:
Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed (ACSIS125)

- working collaboratively to decide how to approach an investigation

LESSON PLAN

Activities	Resources
Activity 1: What Is This? Open the presentation to slide 2. Use the picture of the seesaw to introduce the topic. Briefly discuss some of the features of a seesaw including its: • fulcrum • load • effort 5	• Presentation: Levers
Activity 2: Balance Those Masses! Give out the Balance Those Masses! worksheet to students and allow them to complete the experiment independently. Allow students to discuss their results with the person sitting next to them. Use slides 3 and 4 of the presentation, lead students to conclude that it is possible to balance a heavy mass with a light load but the effort needs to be at a further distance from the fulcrum than that of the load. 20	• Photocopies of the Balance Those Masses! worksheet • pencil, 30 cm ruler, 4 x 50 g masses • Presentation: Levers
Activity 3: Understanding the Lever Give out the Understanding the Lever worksheet to students and ask them to complete it while watching the video. Pause the video to allow students to catch up if necessary. Discuss the 3 types of levers with emphasis on the location of the load, effort and fulcrum. Use the remaining slides of the presentation to revise the difference between the levers. 15	• ClickView video *Levers* **https://clickv.ie./w/lp/19** • Photocopies of the Understanding the Lever worksheet • Presentation: Levers
Activity 4: Find That Lever! Give out the Find That Lever! worksheet and have students find examples of levers in items they use on a regular basis. Allow students to use the Internet if they wish to research on items that cannot be found in the classroom. As a class, discuss the examples of levers found. Students should identify the fulcrum, load and effort when each item is in use. 20	• Photocopies of the Find That Lever! worksheet • Laptops (if students do not have their own, conduct this lesson in a computer lab)

ANSWERS

Balance Those Masses!

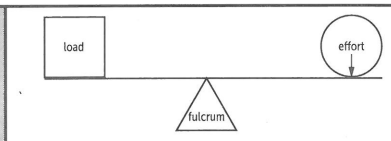
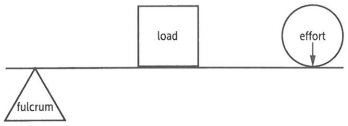
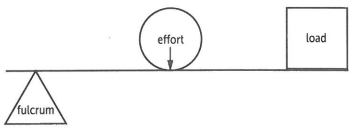
1. Yes
2. Students' answers may vary.
3. Students' answers may vary.
4. We are able to lift a heavy load using a light force.
5. Suggested Answer: The distance between the load and the fulcrum decreases if the load increases.

Understanding the Lever

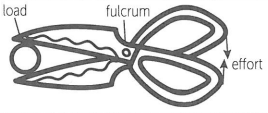
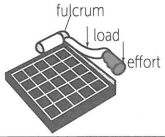
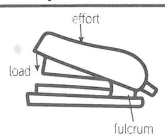
Components of a Lever
Suggested Answer:

1	**Effort**	Force used to move a lever
2	**Load**	The object to be moved
3	**Fulcrum**	The point which a lever pivots

Type of Lever

• Seesaw • Claw hammer • Pliers • Lid openers	
• Wheelbarrow • Nutcracker • Paper guillotine • Bottle opener	
• Forceps • Broom • Fishing rod • Baseball bat	

Find That Lever!
Possible Answer:

Item	Lever Diagram	Type of Lever
Scissors		First class lever
Paper guillotine		Second class lever
Stapler		Third class lever

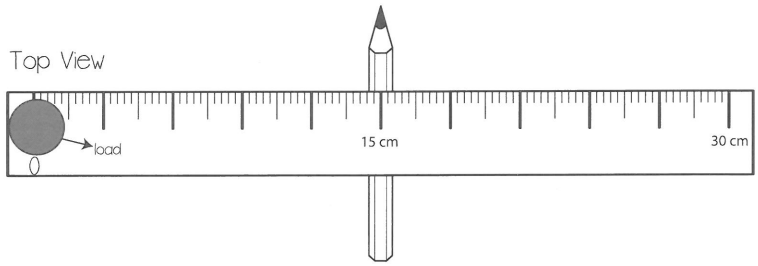
PHYSICAL SCIENCES

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Balance Those Masses!

Materials:

- pencil
- 30 cm ruler
- 4 x 50 g masses



Method:

1. Place a pencil vertically on your desk in front of you.
2. Put a ruler on top of the pencil in a horizontal position. This should lie around the 15 cm mark. Slight adjustments may need to be made depending on the design of the ruler for it to lie perfectly horizontal on the pencil. The pencil is the fulcrum.
3. Add a 50 g mass to the ruler. It should be placed at the far end of the ruler near the 0 cm mark. This is the load.
4. Balance the load by placing another 50 g mass on the opposite end of the ruler. This is the effort.
5. Remove all masses from the ruler.
6. Place 3 x 50 g masses on the 0 cm mark and one 50 g mass on the opposite end of the ruler.
7. Discover a way to balance the load without taking any mass off the ruler. (Hint: You can move the pencil.)
8. Record your findings below.

Questions

1. Were you able to balance the 150 g load using only a 50 g effort? _____
2. When both masses were balanced (neither ends of the ruler touching the table), what was the reading on the ruler where the pencil was located? _____ cm
3. Draw and label a diagram to show how you balanced the load.

4. What is one advantage of using a lever?

_____

_____

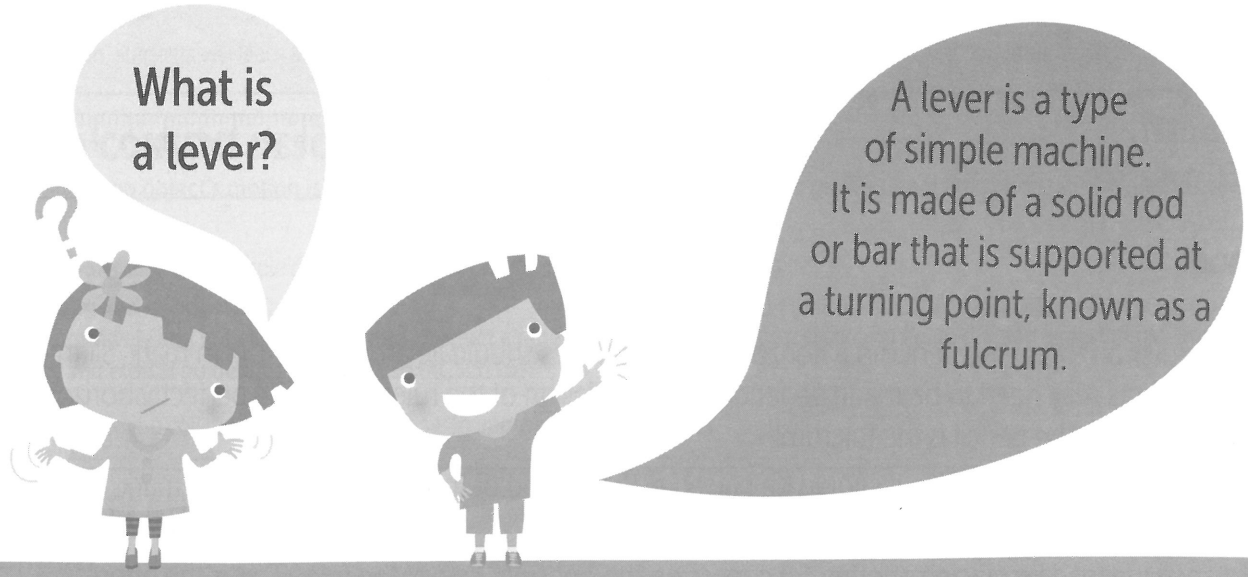
5. Albert conducted the same investigation using 4 x 50 g masses as the load. He found the load could be balanced by the 50 g effort if the pencil was located at the 6 cm mark on the ruler. What does this tell you about the relationship between the mass of the load and its distance from the fulcrum?

_____

_____

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Understanding the Lever



Components of a Lever

A lever consists of 3 components.
Watch the video and write a short definition for each component below.

1	Effort	
2	Load	
3	Fulcrum	

Type of Lever

The type of lever depends on the relative position of the fulcrum, effort and load.
Use the key to draw and label a diagram of each lever below.

List 4 examples of class one levers • • • •	Class one lever diagam
List 4 examples of class two levers • • • •	Class two lever diagam
List 4 examples of class three levers • • • •	Class three lever diagam

Key
- ▲ fulcrum
- ■ load
- ● effort

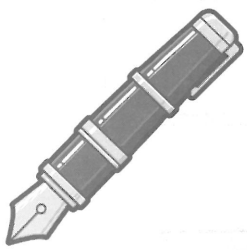
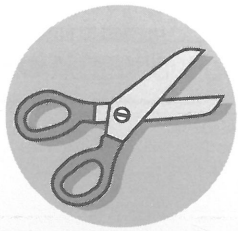
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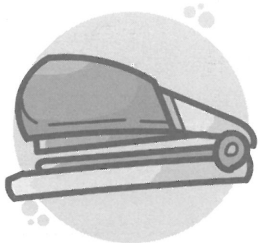
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Levers Are Everywhere!

How many different examples of levers can you find in the classroom?
Draw and label a diagram including the fulcrum, effort and load of each lever you find.
If you are unsure, look up your example on the Internet.

Item	Lever Diagram	Type of Lever



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